



SIMATS ENGINEERING
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Course Code: ITA1409

Slot: A

Course Name: Ethical Hacking

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EXP:08 Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems:

Aim:

To execute and display the results of **Nmap service version detection, aggressive scanning, and operating system detection commands** in Kali Linux.

i. Detect the version of services running

```
(rachitha@kali)-[~]
$ nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 13:11 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000070s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.86 seconds
```

ii. Aggressive scan

```
(rachitha@kali)-[~]
$ sudo nmap -A 127.0.0.1
[sudo] password for rachitha:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 13:19 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000069s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.48 seconds
```

iii. Detect operating system of the target

```
(rachitha@kali)-[~]  
$ sudo nmap -O 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-07 13:22 +0530  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000073s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
Too many fingerprints match this host to give specific OS details  
Network Distance: 0 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds
```

Result:

Thus, the Nmap commands were successfully executed in Kali Linux. The -sV option was used to detect service versions, -A was used to perform an aggressive scan, and -O was used to detect the operating system of the target.